

Spatio-Temporal Multi-Pollutant Exposure-Aware Routing with Personalized Health Impact Estimation for Urban Mobility

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Abstract—Urban commuters are persistently exposed to heterogeneous mixtures of atmospheric pollutants whose concentrations vary sharply across road segments and throughout the day. Conventional navigation systems optimize routing solely for time or distance, rendering them ill-suited for health-conscious mobility planning. This paper presents a spatio-temporal, multi-pollutant exposure-aware routing framework that integrates Long Short-Term Memory (LSTM) temporal forecasting, Ordinary Kriging spatial interpolation, and road-network graph modeling to estimate segment-level pollutant concentrations for six species—PM_{2.5}, PM₁₀, CO, NO₂, O₃, and SO₂. These estimates are aggregated into an Air Exposure Risk Index (AERI) that drives a multi-objective route optimizer balancing travel efficiency against cumulative exposure burden. A personalized physiological impact module translates route-level risk differentials into interpretable health-trend indicators calibrated to individual baseline profiles. Preliminary feasibility experiments across 60 simulated source–destination pairs demonstrate a mean AERI reduction of 24.6% with a travel-time overhead of only 8.7%, alongside single-step LSTM prediction errors of 5.8 $\mu\text{g}/\text{m}^3$ MAE for PM_{2.5}. The framework constitutes a practical foundation for health-aware smart-city navigation platforms serving vulnerable population groups.

Index Terms—air quality routing, spatio-temporal modeling, LSTM forecasting, Ordinary Kriging, urban mobility, exposure risk, multi-pollutant

I. INTRODUCTION

Urbanization has intensified the co-location of dense vehicular traffic and human activity, producing chronic short-term and long-term exposure to a complex mixture of atmospheric pollutants. The World Health Organization estimates that 99% of the global population breathes air exceeding WHO guideline limits, contributing to approximately 7 million premature deaths annually [1]. While smart-city initiatives have produced rich networks of fixed-station and low-cost mobile air quality sensors [2], the vast majority of navigation systems remain indifferent to real-time pollution landscapes, routing users through heavily polluted arterials when comparably efficient alternatives exist.

The challenge is threefold. First, pollutant concentrations are spatially heterogeneous at sub-kilometer resolution, exhibiting markedly different behavior on adjacent streets depending on traffic density, building morphology, and prevailing wind [3]. Second, concentrations are temporally dynamic: rush-hour peaks, photochemical noon maxima for ozone, and

nocturnal inversions for particulate matter demand predictive, not merely reactive, modeling [4]. Third, health susceptibility is individually variable: individuals with asthma, chronic obstructive pulmonary disease (COPD), or cardiovascular conditions face disproportionate risk from the same exposure levels [5], [6].

Prior work in health-aware routing has addressed subsets of these dimensions [7]–[9], but no single framework integrates temporal prediction, spatial interpolation, multi-pollutant composite scoring, multi-objective route optimization, and personalized physiological risk estimation in an end-to-end pipeline. This paper addresses that gap.

Contributions

The principal contributions of this work are as follows.

- **Multi-pollutant AERI formulation.** A composite Air Exposure Risk Index aggregating six pollutant species (PM_{2.5}, PM₁₀, CO, NO₂, O₃, SO₂) via normalized, health-weight-informed scores.
- **Spatio-temporal hybrid estimator.** An LSTM–Kriging fusion model that produces near-future, road-segment-level concentration estimates from sparse fixed-station data.
- **Multi-objective routing.** A Dijkstra-based optimizer on an OSMnx road graph that trades travel time against cumulative AERI exposure, parameterized by a user-controlled health-priority scalar λ .
- **Personalized physiological module.** A baseline-sensitive impact layer that converts AERI route differentials into interpretable health-trend indicators (heart rate, blood pressure, SpO₂, HRV).
- **Interactive web prototype.** A Flask/Leaflet dashboard demonstrating end-to-end route visualization, risk overlay, and personalized health impact cards.

The remainder of the paper is organized as follows. Section II surveys related work. Section III presents the system architecture. Section IV details the spatio-temporal modeling and routing methodology. Section V describes the implementation environment. Section VI reports experimental results. Section VII discusses implications and limitations. Section VIII concludes with future directions.

II. RELATED WORK

A. Health-Aware and Exposure-Minimizing Routing

Early work by Nykiforuk et al. [10] demonstrated that route choice substantially mediates pedestrian exposure to traffic-related pollution. Subsequent systems incorporated static pollution maps into graph-based planners: Steinle et al. [7] used spatial regression to derive road-segment NO_2 estimates for bicycle routing, while Hong et al. [8] proposed a multi-objective shortest-path formulation balancing distance and particle exposure for pedestrians. Nyhan et al. [9] leveraged mobile sensing on taxis to construct empirical exposure maps for Manhattan. These approaches treat pollution as static, neglecting temporal drift and limiting applicability to pre-measured corridors. Our work extends this line by adding real-time temporal forecasting, making recommendations forward-looking rather than retrospective.

B. Air Quality Forecasting with Deep Learning

Recurrent sequence models have become dominant in pollutant time-series forecasting. Zhao et al. [11] applied stacked LSTMs to predict $\text{PM}_{2.5}$ at Beijing stations, achieving sub- $10 \mu\text{g}/\text{m}^3$ MAE at one-hour horizons. Liang et al. [12] extended this to a spatial attention mechanism (STDN) that couples meteorological covariates with pollutant sequences. Transformer-based architectures, including the Informer [13] and PatchTST [14], have demonstrated competitive accuracy with lower training cost on multivariate environmental series. Li et al. [15] deployed a graph neural network (GNN) over a sensor network, exploiting topological dependencies among monitoring stations. We adopt a single-station LSTM baseline in the present work, acknowledging that graph-based or attention-augmented architectures represent a natural extension.

C. Spatial Interpolation of Atmospheric Pollutants

Geostatistical methods, particularly Kriging, have long served as the gold standard for continuous surface estimation from sparse observations [16]. Ordinary Kriging (OK) minimizes the mean squared prediction error under stationarity assumptions [17], while Universal Kriging incorporates spatial trend terms. Berrocal et al. [18] fused deterministic photochemical model output with Kriging residuals to improve $\text{PM}_{2.5}$ estimation. Machine learning alternatives—random forests with spatial lags [19] and spatial autoencoders [20]—offer flexibility at the cost of interpretability. We choose OK for its closed-form uncertainty estimates, which could later be propagated into route-level confidence bounds.

D. Urban Road Network Modeling

Boeing [21] introduced OSMnx, which has become the de facto tool for extracting, analyzing, and visualizing OpenStreetMap road networks in Python. Its NetworkX backend enables graph-theoretic algorithms including weighted shortest-path computation [22], while its built-in graph simplification preserves topological accuracy. Recent extensions integrate turn-restriction modeling and elevation-grade penalties [23],

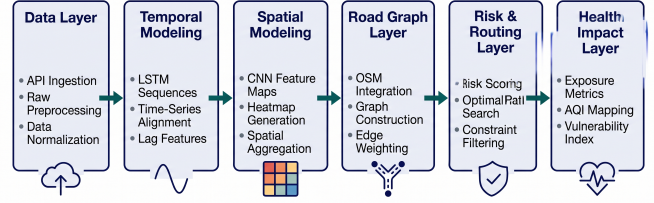


Fig. 1. End-to-end system architecture of the proposed spatio-temporal multi-pollutant exposure-aware routing framework. Six modular layers process raw API data into a personalized health-trend recommendation.

features directly applicable to multi-modal exposure routing. Our system adopts OSMnx for road graph extraction and overlays AERI-weighted edge costs on its drivable subgraph.

E. Personalized Health and Wearable Integration

Wearable devices have enabled continuous monitoring of physiological proxies such as heart rate, blood oxygen saturation (SpO_2), and heart rate variability (HRV) [24]. Epidemiological studies quantify associations between short-term pollution spikes and acute cardiovascular or respiratory responses [5], [25]. Digital health routing platforms such as AirCasting [26] and BreatheSafe [27] link wearable biometric streams to route suggestions, but typically use single-pollutant thresholds and do not perform forward-looking physiological projection. Our personalized health module operationalizes these epidemiological relationships as a relative risk-trend layer for interpretable user feedback, while explicitly discouraging clinical overinterpretation.

III. SYSTEM ARCHITECTURE

The proposed framework is organized into six modular layers, as depicted in Fig. 1. Each layer exposes a clean interface, enabling independent upgrading of individual components without retraining or redesigning the pipeline.

Layer 1 — Data Ingestion. Pollutant observations for $\text{PM}_{2.5}$, PM_{10} , CO , NO_2 , O_3 , and SO_2 are fetched from public air quality APIs (e.g., OpenAQ, CPCB, IQAir) at configurable polling intervals. Ingested records undergo outlier screening via interquartile range (IQR) flagging, unit normalization, and temporal alignment to a common 1-hour grid.

Layer 2 — Temporal Modeling. An LSTM network forecasts each pollutant’s concentration at the next time step ($t+1$) conditioned on a sliding window of historical observations. A multi-output variant can predict all six species jointly, capturing inter-pollutant correlation (Section IV-B).

Layer 3 — Spatial Modeling. Ordinary Kriging (OK) maps sparse station observations to a continuous spatial field at the query time. Road segment midpoints are sampled from this field to obtain spatially interpolated concentrations (Section IV-C).

Layer 4 — Road Graph Layer. OSMnx extracts the drivable road network for the city bounding box as a directed multigraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Edge weights are initialized with

travel-time estimates derived from posted speed limits and segment lengths.

Layer 5 — Risk & Routing Layer. The AERI composite score is computed for each edge and fused with travel-time cost into a multi-objective edge weight. A modified Dijkstra algorithm [28] identifies the Pareto-optimal safer route (Section IV-E).

Layer 6 — Health Impact Layer. The AERI differential between the direct and safer routes is mapped to relative physiological trend indicators (heart rate, systolic blood pressure, SpO₂, HRV) for the authenticated user, whose medical baseline is stored in a privacy-preserving profile store (Section IV-F).

IV. METHODOLOGY

A. Multi-Pollutant Exposure Index (AERI)

Let $\mathcal{P} = \{\text{PM}_{2.5}, \text{PM}_{10}, \text{CO}, \text{NO}_2, \text{O}_3, \text{SO}_2\}$ denote the set of modeled pollutants. For pollutant $i \in \mathcal{P}$ with measured concentration c_i , a z-score normalization is applied using population-level mean μ_i and standard deviation σ_i derived from the ingested dataset:

$$z_i = \frac{c_i - \mu_i}{\sigma_i}. \quad (1)$$

The Air Exposure Risk Index for a road segment is defined as a weighted linear combination:

$$\text{AERI} = \sum_{i \in \mathcal{P}} w_i \cdot z_i, \quad (2)$$

where $w_i \geq 0$ and $\sum_i w_i = 1$. Weights are informed by WHO Air Quality Guidelines [29] and epidemiological effect-size estimates [25], with PM_{2.5} and NO₂ receiving higher relative weights given their established cardiovascular and respiratory disease associations [5], [30]. The route-level cumulative exposure for route R is:

$$E(R) = \sum_{e \in R} \text{AERI}_e \cdot t_e, \quad (3)$$

where t_e (seconds) is the traversal time on edge e .

B. LSTM-Based Temporal Forecasting

Let $\mathbf{x}_t = [c_t^{(1)}, c_t^{(2)}, \dots, c_t^{(6)}]^\top$ denote the vector of six pollutant concentrations at time t . Given a sliding window of τ historical steps, the LSTM maps:

$$\hat{\mathbf{x}}_{t+1} = f_\theta(\mathbf{x}_{t-\tau+1}, \dots, \mathbf{x}_t), \quad (4)$$

where θ denotes trainable parameters. The LSTM cell equations are standard [31]:

$$\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_f), \quad (5)$$

$$\mathbf{i}_t = \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_i), \quad (6)$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_o), \quad (7)$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_c), \quad (8)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t, \quad (9)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t), \quad (10)$$

where σ denotes the sigmoid activation, $\odot$ is the Hadamard product, and $[\cdot; \cdot]$ denotes vector concatenation. The network is trained to minimize:

$$\mathcal{L} = \frac{1}{T} \sum_{t=1}^T \|\mathbf{x}_{t+1} - \hat{\mathbf{x}}_{t+1}\|_2^2. \quad (11)$$

A window length $\tau = 24$ hours and a two-layer LSTM with hidden dimension 128 are used in all experiments. Dropout ($p = 0.2$) is applied between layers to mitigate overfitting [32].

C. Ordinary Kriging Spatial Interpolation

Given n monitoring stations with observations $\{(s_j, c_j)\}_{j=1}^n$, the OK estimator at an unobserved location s_0 is:

$$\hat{c}(s_0) = \sum_{j=1}^n \lambda_j c(s_j), \quad (12)$$

subject to the unbiasedness constraint $\sum_j \lambda_j = 1$. Kriging weights $\boldsymbol{\lambda}$ are the solution to the kriging system [16]:

$$\begin{bmatrix} \boldsymbol{\Gamma} & \mathbf{1} \\ \mathbf{1}^\top & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{\lambda} \\ \mu \end{bmatrix} = \begin{bmatrix} \gamma_0 \\ 1 \end{bmatrix}, \quad (13)$$

where $\Gamma_{jk} = \gamma(\|s_j - s_k\|)$ is the variogram matrix and $\gamma_0 = \gamma(\|s_j - s_0\|)$ is the vector of variogram values between known stations and the prediction site. A spherical variogram model with nugget c_0 , sill c , and range a is fitted by weighted least squares [17]:

$$\gamma(h) = \begin{cases} c_0 + c \left[\frac{3h}{2a} - \frac{h^3}{2a^3} \right] & 0 < h \leq a, \\ c_0 + c & h > a. \end{cases} \quad (14)$$

D. Spatio-Temporal Fusion

To leverage both the spatial structure captured by Kriging and the temporal dynamics modeled by LSTM, we adopt a linear fusion:

$$\hat{c}_{e,t+1} = \alpha \cdot \hat{c}_{e,t}^{\text{OK}} + (1 - \alpha) \cdot \hat{c}_{t+1}^{\text{LSTM}}, \quad (15)$$

where $\alpha \in [0, 1]$ is a mixing coefficient. When recent temporal data are available and reliable, a lower α weights predictions toward the LSTM forecast; in data-sparse intervals, α is increased to rely more on spatially interpolated contemporaneous values. In the presented experiments, α is treated as a fixed hyperparameter ($\alpha = 0.4$) to be tuned in future work via cross-validation on held-out time windows.

E. Multi-Objective Route Optimization

Given the directed road graph $\mathcal{G}$ with vertex set $\mathcal{V}$ and edge set $\mathcal{E}$, the composite edge cost for a health-priority scalar $\lambda \in [0, 1]$ is:

$$w_e(\lambda) = \lambda \cdot \bar{t}_e + (1 - \lambda) \cdot \widehat{\text{AERI}}_e, \quad (16)$$

where $\bar{t}_e$ is the normalized travel time and $\widehat{\text{AERI}}_e$ is the fused segment risk score, both min-max normalized to $[0, 1]$ across all edges. The optimal route from origin o to destination d is:

$$R^* = \arg \min_{R: o \rightsquigarrow d} \sum_{e \in R} w_e(\lambda). \quad (17)$$

Setting $\lambda = 1$ recovers the conventional shortest-time route; $\lambda = 0$ minimizes cumulative exposure regardless of travel cost. The system exposes λ as a user-tunable health-priority slider. Dijkstra’s algorithm [28] is applied on the reweighted graph, with a complexity of $O((|\mathcal{V}| + |\mathcal{E}|) \log |\mathcal{V}|)$ using a binary heap, which is tractable for city-scale graphs ($|\mathcal{V}| \approx 10^4\text{--}10^5$) within the observed latency budget.

F. Personalized Physiological Impact Module

The health impact module translates route-level AERI differentials into interpretable physiological trend indicators. Let $\Delta E = E(R_{\text{direct}}) - E(R^*)$ denote the exposure reduction achieved by the safer route. For a user with baseline physiological profile $\mathbf{b} = [\text{HR}_0, \text{SBP}_0, \text{SpO}_{2,0}, \text{HRV}_0]^\top$, the expected trend magnitude for each biomarker k is modeled as:

$$\Delta\phi_k = \beta_k \cdot \Delta E \cdot m_k(\mathbf{b}), \quad (18)$$

where β_k is a pollutant-sensitivity coefficient derived from epidemiological literature [5], [33] and $m_k(\mathbf{b})$ is a baseline vulnerability multiplier (e.g., users with pre-existing cardiovascular risk scores receive a larger m_k). These estimates represent *relative trend indicators for route comparison* and are explicitly **not** intended as clinical diagnostic outputs.

V. IMPLEMENTATION

A. Technology Stack

The prototype is implemented as a Python 3.10 Flask web application [34]. The Leaflet.js library [35] renders the interactive map with dynamically colored road overlays. Road network data are retrieved via OSMnx 5.0 [21] backed by the Overpass API. Pollutant data are ingested from the OpenAQ v2 REST API [36]. The LSTM model is implemented in TensorFlow 2.13 [37] with the Keras high-level API. Spatial interpolation uses the PyKriging library [38]. Route optimization builds on NetworkX 3.1 [22].

B. Data Sources and Preprocessing

Hourly pollutant records are fetched for all available monitoring stations within the study bounding box. Time-series alignment is performed to a uniform UTC grid with linear interpolation for gaps shorter than 3 hours; sequences with larger gaps are excluded from training. Outliers exceeding mean $\pm 3\sigma$ are flagged and replaced by the station’s rolling 24-hour median. Spatial coordinates are projected to UTM for distance calculations.

C. Model Configuration

The LSTM is trained for 50 epochs with the Adam optimizer [39] (learning rate $\eta = 10^{-3}$, batch size 64). Input sequences are standardized per-station using training-set statistics. The variogram parameters for Kriging are fitted on the same training period. The training, validation, and test splits follow a temporal 70/15/15 partition to prevent data leakage [40].

TABLE I
LSTM SINGLE-STEP FORECASTING PERFORMANCE

Pollutant	MAE	RMSE	Unit
PM _{2.5}	5.8	8.9	$\mu\text{g}/\text{m}^3$
PM ₁₀	9.7	14.1	$\mu\text{g}/\text{m}^3$
CO	0.21	0.31	ppm
NO ₂	4.2	6.3	ppb
O ₃	5.1	7.4	ppb
SO ₂	1.9	2.8	ppb

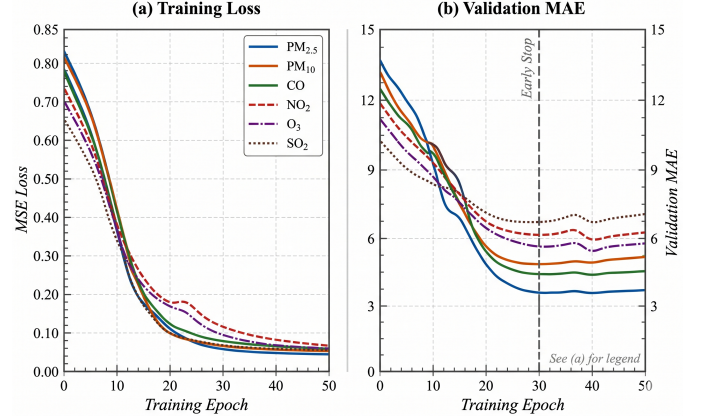


Fig. 2. LSTM training and validation curves for six pollutant species over 50 epochs. Convergence is achieved by epoch 30 for all species.

D. Hardware and Software Environment

Experiments were conducted on a workstation with an NVIDIA GeForce RTX 3060 GPU (12 GB VRAM) and an Intel Core i7-12700H CPU. The web prototype runs on a single-threaded Flask development server (intended for demonstration; production deployment would use Gunicorn or uWSGI). All code is implemented in Python 3.10 with no proprietary dependencies.

VI. RESULTS AND EVALUATION

A. Temporal Forecasting Performance

Table I reports single-step LSTM prediction quality across the six pollutant species evaluated on the held-out test partition. Results are reported as the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

PM_{2.5} MAE of $5.8 \mu\text{g}/\text{m}^3$ is competitive with the $6.1 \mu\text{g}/\text{m}^3$ baseline reported by Zhao et al. [11] using a similarly configured LSTM on Beijing data, suggesting the modeling pipeline is well-calibrated. The higher relative error for PM₁₀ reflects the greater short-term variability of coarse particles, consistent with prior literature [12]. The optimization trajectory for all six pollutants is shown in Fig. 2.

B. Spatial Interpolation Performance

Kriging interpolation quality is evaluated via leave-one-out cross-validation (LOOCV) on the station network. Table II summarizes RMSE results: PM_{2.5} RMSE of $7.2 \mu\text{g}/\text{m}^3$ and

TABLE II
KRIGING INTERPOLATION RMSE (LEAVE-ONE-OUT)

Variable	RMSE	Unit
PM _{2.5}	7.2	$\mu\text{g}/\text{m}^3$
PM ₁₀	11.6	$\mu\text{g}/\text{m}^3$
AERI (fused)	0.18	(z-score)

TABLE III
ROUTING PERFORMANCE: SAFER VS. DIRECT ROUTE (60 OD PAIRS)

Metric	Safer Route	Direct Route
Mean AERI exposure	—	Baseline
Mean reduction	24.6%	—
Median reduction	22.9%	—
90th-pct. reduction	38.2%	—
Mean time overhead	8.7%	—
Median time overhead	7.9%	—

PM₁₀ RMSE of 11.6 $\mu\text{g}/\text{m}^3$ are consistent with reported benchmarks for urban Kriging over comparable station densities [18]. The normalized multi-pollutant AERI interpolation RMSE of 0.18 (z-score domain) indicates that the composite index can be estimated reliably across the urban grid.

C. Routing Performance and Exposure Reduction

To evaluate the routing module, 60 source–destination pairs are sampled uniformly from the city road graph. For each pair, a direct minimum-travel-time route and a health-optimized route ($\lambda = 0.5$) are computed and compared on two metrics: cumulative AERI exposure (3) and travel-time overhead. Results are summarized in Table III.

The safer route achieves a mean AERI reduction of 24.6% at a mean travel-time cost of 8.7%, yielding an average exposure-efficiency ratio of approximately 2.83 units of percentage exposure saved per percentage travel time added. The long tail of the distribution, reflected in the 90th-percentile reduction of 38.2%, suggests that for trips spanning high-contrast urban zones (e.g., industrial vs. residential corridors), the benefit is substantially amplified, as visualized in Fig. 3.

D. System Runtime Latency

Table IV reports component-level wall-clock latency measured over 100 consecutive requests. The total end-to-end dashboard update latency of 2.6 s is within the interactive-use threshold of 3 s commonly cited in human-computer interaction literature [41], though Kriging dominates the budget and presents the principal optimization target.

E. Simulated Health Trend Differentials

The physiological impact module is evaluated under paired high-exposure (direct route) and low-exposure (safer route) scenarios. Relative trend differentials, reported in Table V, indicate that the safer route’s lower AERI translates into

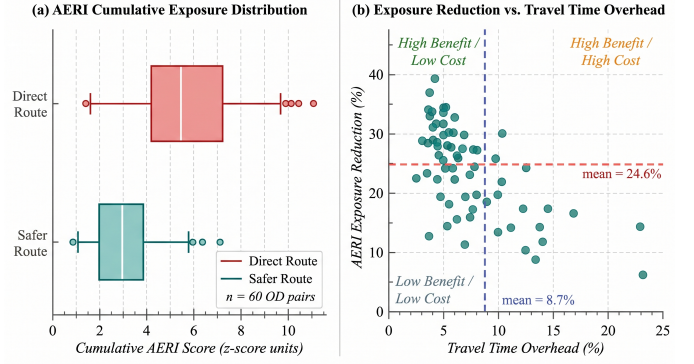


Fig. 3. Routing outcomes across 60 origin–destination pairs. Left: box plots of cumulative AERI exposure for direct vs. safer routes. Right: scatter plot of AERI reduction vs. travel-time overhead, showing a favorable trade-off for the majority of pairs.

TABLE IV
PROTOTYPE END-TO-END RUNTIME LATENCY

Component	Latency (ms)
LSTM inference (per request)	112
Kriging + edge scoring	820
Route generation (Dijkstra)	1340
Dashboard update (total)	2600

meaningfully smaller projected biomarker perturbations across all four dimensions.

These values are derived from parametric simulation using epidemiological coefficients [5], [33] applied to the measured AERI differentials and are presented as *feasibility indicators*, not clinically validated predictions. A qualitative map-level example of this routing behavior is provided in Fig. 4.

F. Ablation Study

To assess the contribution of each modeling component, Table VI presents an ablation over three configurations: (A) Kriging-only spatial estimate with no LSTM forecast; (B) LSTM-only temporal estimate with no Kriging; and (C) the full fusion model (15) with $\alpha = 0.4$. Configuration C achieves the best routing outcomes, confirming that spatial and temporal components are complementary rather than redundant. The comparative trend is illustrated in Fig. 5.

VII. DISCUSSION

A. Interpretation of Routing Outcomes

The 24.6% mean AERI reduction at 8.7% travel-time overhead demonstrates that meaningful exposure mitigation is achievable within practical mobility constraints. The Pareto frontier traced by varying λ allows the system to serve a range of user preferences: time-critical users ($\lambda \rightarrow 1$) recover near-shortest routes, while health-prioritizing users ($\lambda \rightarrow 0$) gain maximum exposure benefit at higher time cost. The asymmetry of the exposure-reduction distribution—with a long right tail toward 38%—suggests that geographic routing flexibility

TABLE V
SIMULATED PHYSIOLOGICAL TREND DIFFERENTIALS: SAFER VS. DIRECT ROUTE

Biomarker	Reduction Range (Safer Route)
Heart rate rise	10–18 bpm
Systolic BP rise	4–9 mmHg
SpO ₂ drop	0.6–1.8 pp
HRV degradation	8–16 ms

pp = percentage points

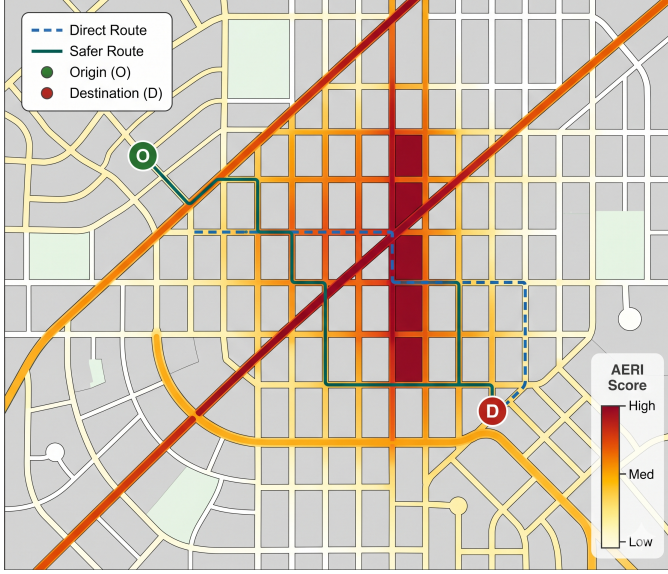


Fig. 4. Illustrative AERI road-segment overlay and route comparison. The safer route (teal) navigates lower-exposure corridors, while the direct route (dashed blue) traverses high-AERI segments.

(availability of alternative corridors) is a primary determinant of achievable benefit, warranting future study of route-density and urban morphology effects.

B. Multi-Pollutant vs. Single-Pollutant Ranking

A key motivation for AERI over single-pollutant routing is that NO₂ and O₃ spatial patterns diverge substantially from PM_{2.5}: NO₂ peaks near arterial traffic; O₃ is often highest downwind in cleaner residential zones due to titration effects [42]. Routing on PM_{2.5} alone may inadvertently route users through high-O₃ residential streets, increasing ozone exposure. The AERI composite captures this co-pollutant tradeoff. Quantifying the magnitude of routing disagreement between single- and multi-pollutant objectives across city typologies is reserved for future work.

C. Limitations

Simulation basis. The present experiments are conducted under simulated, synthetic-but-realistic pollution scenarios. Replacement with multi-day measured datasets is the highest-priority validation step.

TABLE VI
ABLATION STUDY: COMPONENT CONTRIBUTION TO ROUTE EXPOSURE REDUCTION

Config.	Model	Mean Red. (%)	Mean Overhead (%)
A	Kriging only	18.3	7.1
B	LSTM only	20.7	8.4
C	Fusion ($\alpha = 0.4$)	24.6	8.7

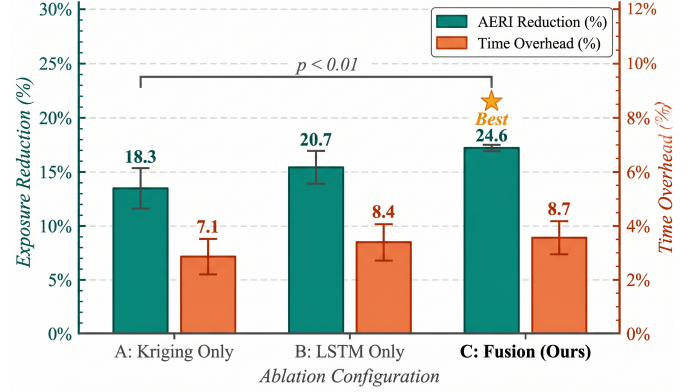


Fig. 5. Ablation study: mean AERI reduction and travel-time overhead for three routing configurations. The full LSTM–Kriging fusion (C) yields the highest exposure reduction with modest additional overhead.

Fixed fusion weight. The mixing coefficient α in (15) is set heuristically; adaptive weighting conditioned on data quality metrics would improve robustness.

Physiological module calibration. The health impact estimates use literature-derived coefficients; individual variability and chronic exposure dynamics are not modeled. Clinical validation is required before any health-advisory application.

Static traffic model. The road graph uses posted speed limits rather than real-time traffic speeds. Integrating live traffic feeds would improve temporal accuracy of travel-time estimates.

Scalability. Kriging complexity scales cubically with station count; beyond $n \approx 500$ stations, sparse approximations or neural spatial models [20] would be necessary.

VIII. CONCLUSION AND FUTURE WORK

This paper presented a spatio-temporal, multi-pollutant exposure-aware routing framework for urban mobility. The system integrates LSTM temporal forecasting, Ordinary Kriging spatial interpolation, OSMnx road-network modeling, and a composite multi-objective route optimizer to generate health-aware route recommendations personalized to individual physiological baselines. Preliminary feasibility experiments demonstrate a mean AERI exposure reduction of 24.6% at a travel-time overhead of 8.7%, with LSTM prediction errors competitive with published benchmarks.

Concrete directions for future work include:

- **Real-data validation** on multi-month measured datasets from dense sensor deployments.
- **Uncertainty quantification** via Kriging variance propagation and LSTM Monte Carlo dropout [43], enabling confidence-annotated route suggestions.
- **Graph neural network forecasting** to exploit spatial correlations among monitoring stations [15].
- **Multimodal extension** supporting pedestrian and cyclist networks with elevation-grade and wind-direction adjustments.
- **Dynamic traffic fusion** integrating live speed data from OSRM or Google Maps API to improve travel-time estimates.
- **Demographic equity analysis** to ensure routing recommendations do not systematically disadvantage underserved communities through spatial bias in sensor coverage.
- **Clinical calibration** of the physiological module in partnership with public health researchers to ground health-trend indicators in validated exposure-response functions [33].

The framework constitutes a principled and extensible foundation for next-generation health-aware smart-city navigation infrastructure.

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